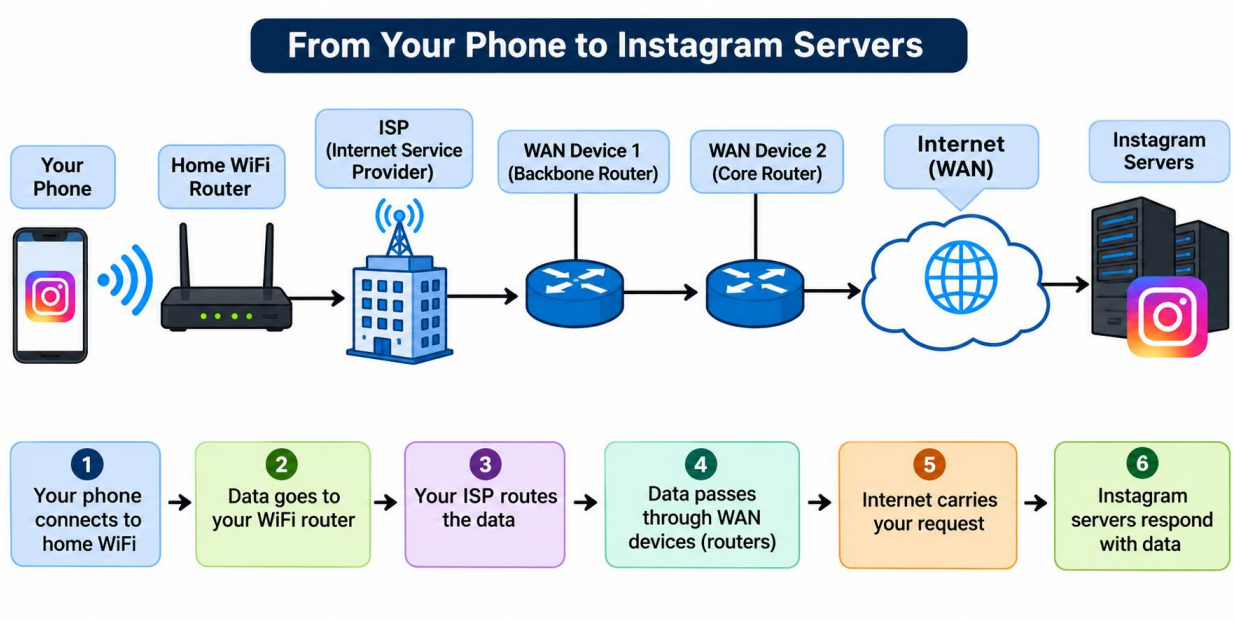
Wide Area Networks (WAN)

Task 1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Answer:



How the Connection Works:

1. The phone connects to the Home Wi-Fi Router wirelessly using Wi-Fi.

2. The Home Wi-Fi Router sends the data to the ISP (Internet Service Provider).

3. The ISP routes the data to WAN Device 1 (Backbone Router).

4. The data then passes through WAN Device 2 (Core Router) towards the Internet.

5. The Internet carries the request to Instagram Servers.

6. Instagram Servers process the request and send photos, videos, and reels back to the phone through the same network path.

Note: The diagram shows two WAN devices: Backbone Router and Core Router.

Task 2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Answer:

WAN technology	Speed	Cost	Where you might see it
Broadband Internet (fibre / DSL / cable)	Medium to high; about 50 Mbps to 1 Gbps. Shared with other users, so speed can vary.	Low. Home plans are about a few hundred to a couple of thousand rupees per month.	Your home Wi-Fi, small shops and small offices.
Leased Line	High and steady; dedicated line, same speed for upload and download (from 2 Mbps up to 10 Gbps or more).	High. A dedicated line with a guaranteed speed costs much more than broadband; the price depends on speed and distance.	A company office such as the Zomato office, colleges, banks and data centres that need a stable internet connection.
MPLS (Multiprotocol Label Switching)	High and reliable; usually 2 Mbps to 1 Gbps per site, with priority for important traffic (QoS).	High. It is a private service provided by a telecom company, and it costs more than broadband.	Large companies with many branches, such as banks, and big organizations like Flipkart or IRCTC-type systems that need a private, safe link between sites.

Task 3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Answer:

A VPN builds a secure, private connection (a "tunnel") across a public WAN, which is the internet, so that your data is protected even when you use an unsafe network like coffee shop Wi-Fi.

1. You connect your phone or laptop to the coffee shop Wi-Fi. This Wi-Fi is public, so other people on it could try to watch the traffic.
2. You turn on the VPN app. It connects to a VPN server somewhere on the internet (the WAN) and creates an encrypted tunnel between your device and that server.
3. When you open Spotify or YouTube, your data is encrypted on your device first and then goes through the tunnel. Anyone on the coffee shop Wi-Fi can see only unreadable, scrambled data going to the VPN server.
4. The VPN server decrypts the data and sends your request to Spotify or YouTube over the normal internet. The reply comes back to the VPN server, is encrypted again, and is sent back through the tunnel to your device.
5. Spotify or YouTube sees the VPN server's IP address instead of your real IP address, which also adds some privacy.

So the VPN does not create a new network. It uses the existing public WAN (the internet) but wraps the data in encryption, which keeps it private and safe from attackers on the same Wi-Fi.

Task 4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Answer:

Suggested technology: MPLS (a private MPLS VPN service from a telecom provider).

Reason 1: Reliable, private and prioritized. MPLS runs on the provider's private network and not on the public internet, so it is more secure and has less delay and packet loss. It also gives priority (QoS) to important traffic like order, stock and billing data, and the provider guarantees a service level (SLA). This is important for warehouses that cannot stop working.

Reason 2: Easy to connect many sites as one network. MPLS joins all the warehouses (Ahmedabad, Mumbai and Delhi) into one private network where any site can talk directly to any other site, and a new warehouse can be added easily by the provider. A dedicated leased line would need a separate line between each pair of sites, which is costly, and broadband alone is shared and less predictable.

Note: Broadband with a VPN is cheaper, but for large warehouses that depend on constant, secure data flow, MPLS is the better choice.